The relation between (8.10)–(8.13) and (8.2)–(8.4) becomes evident through the substitution $\boldsymbol{p} = \tilde{\boldsymbol{p}}/s$ and $\xi = p_s/\mu$. Then (8.10)–(8.11) are seen to be the same as the corresponding equations in the Nosé-Hoover method, and the final pair of equations become

$$\frac{d}{dt}s = s\xi,$$
$$\frac{d}{dt}\xi = \frac{1}{\mu}\left(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N}_d\mathrm{k_B T} - \Delta H_{\mathrm{N}}\right).$$

The only difference between the last equation and (8.4) lies in the perturbation $\mu^{-1}\Delta H_{\mathrm{N}}$, which is zero along the exact solution. Under discretization, however, we do not expect ΔH_{N} to vanish, so trajectories obtained from Nosé-Poincaré and Nosé-Hoover will differ considerably over long time simulations.

A benefit of respecting the symplectic structure lies in the prospect of using backward error analysis to analyze the behavior of the method. As shown in [40] it is possible to work out the terms of the perturbative expansion of the Nosé-Poincaré Hamiltonian, i.e. to calculate the expansion

$$\tilde{H}_h = H_{\mathrm{NP}} + h^2 H_{[2]} + \mathcal{O}(h^4)$$

for a second order, symmetric, symplectic numerical method. Then one may determine the effective distribution associated to the symplectic discretization method by "integrating out" the microcanonical density with respect to the auxiliary variable and its conjugate momentum. One obtains in this way

$$\tilde{\rho}\,(\boldsymbol{q},\boldsymbol{p}) = \rho_\beta(\boldsymbol{q},\boldsymbol{p})\zeta(\boldsymbol{q},\boldsymbol{p},h) + \mathcal{O}\left(h^4\right).$$

An explicit formula for ζ is given in [40] for several symplectic methods. During simulation, one may calculate the factor ζ and use this quantity to adjust averages. For example, if we wish to calculate the average of a function $g(\boldsymbol{q},\boldsymbol{p})$, we may perform the calculation using ν steps of the Nosé-Poincaré method with the given symplectic, time-reversible, second order integrator, calculate g_n and ζ_n at each timestep, and approximate the canonical average from

$$\langle g\rangle = \frac{1}{\nu+1}\sum_{n=0}^{\nu} g_n\zeta_n^{-1} + \mathcal{O}(h^4).$$

For further details, refer to [40]. Technically, a similar calculation can be carried out for the Nosé-Hoover formulation, but the calculations are more involved.

A good symplectic integrator for Nosé-Poincaré has been given by Nosé [287]. For further discussion and numerical experiments using this idea, see [94] (which also considers stochastic methods).